FROM STREAMS TO STAGES

To what extent do digital streaming metrics predict the physical reality of global concert tours?

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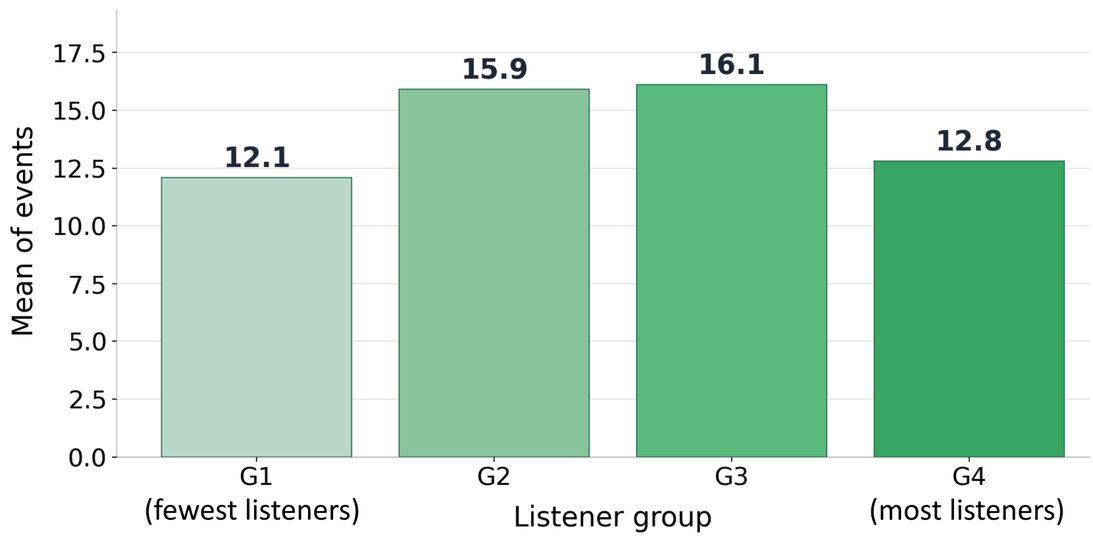
Introduction

The music industry is increasingly shaped by digital streaming platforms and global online audiences. This project investigates how streaming popularity relates to real-world concert activity. Using data from streaming platforms and ticketing services, we analyze the relationship between listener numbers, geographic demand, and concert scheduling.

Streaming & Ticket Power

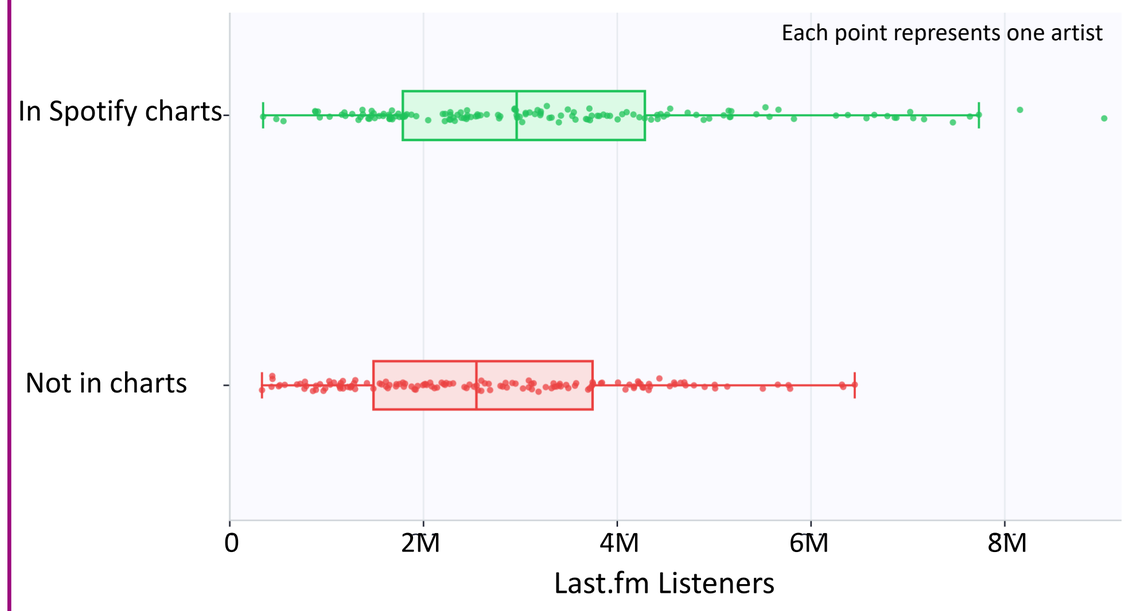
How strongly is digital popularity on Last.fm related to the scale and intensity of touring?

Listeners vs. Tour Size



Artists with more Last.fm listeners tend to schedule slightly more events on average, indicating a weak positive relationship between popularity and tour size. However, the pattern is not fully consistent, as the largest listener group does not continue to increase. This suggests that listener count alone does not determine tour size.

Spotify Chart vs. Non-Chart

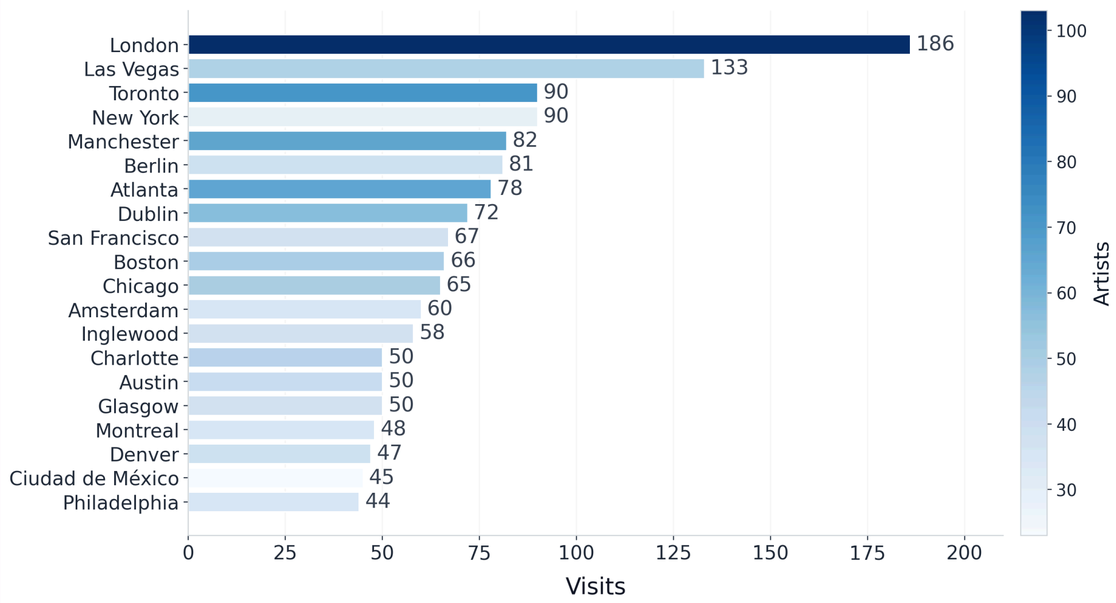


Artists appearing in Spotify charts show clearly higher listener counts compared to non-chart artists. This highlights a strong difference in overall popularity between the two groups. Chart presence therefore serves as a clear indicator of broader audience reach.

Geographic Analyses

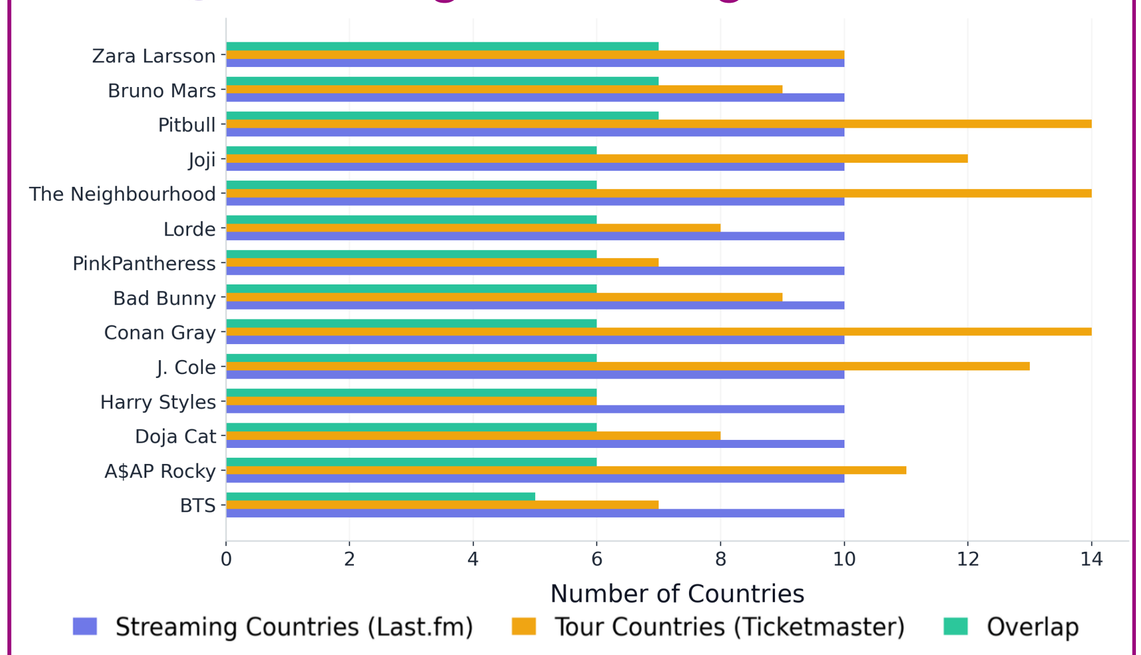
How do artists choose where to tour - do they revisit cities, play in capitals, and follow their listeners?

Most Visited Cities



Touring is focused on a small number of major cities, which are visited much more often than others. These cities likely represent important markets where artists expect strong and reliable demand. This suggests that concerts are not spread evenly, but concentrated in well-established locations.

Streaming vs. Tour Alignment

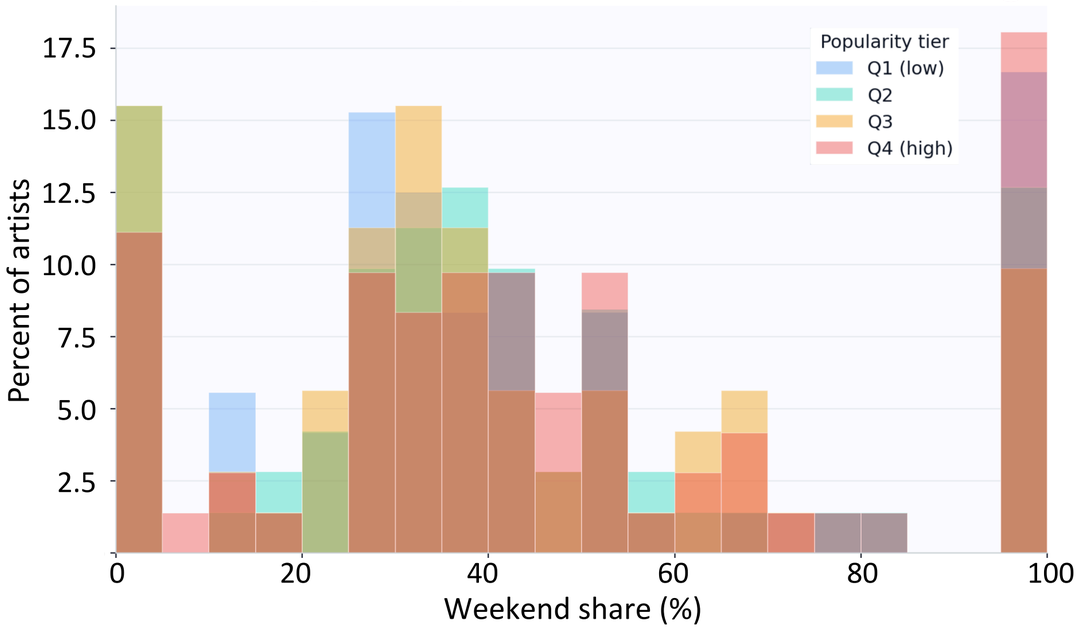


While some artists perform in many of the countries where they have listeners, others show a clear gap between streaming presence and tour locations. This indicates that touring does not always fully reflect global audience distribution. The chart above displays the best aligned artists in the dataset.

Market Time & Scheduling

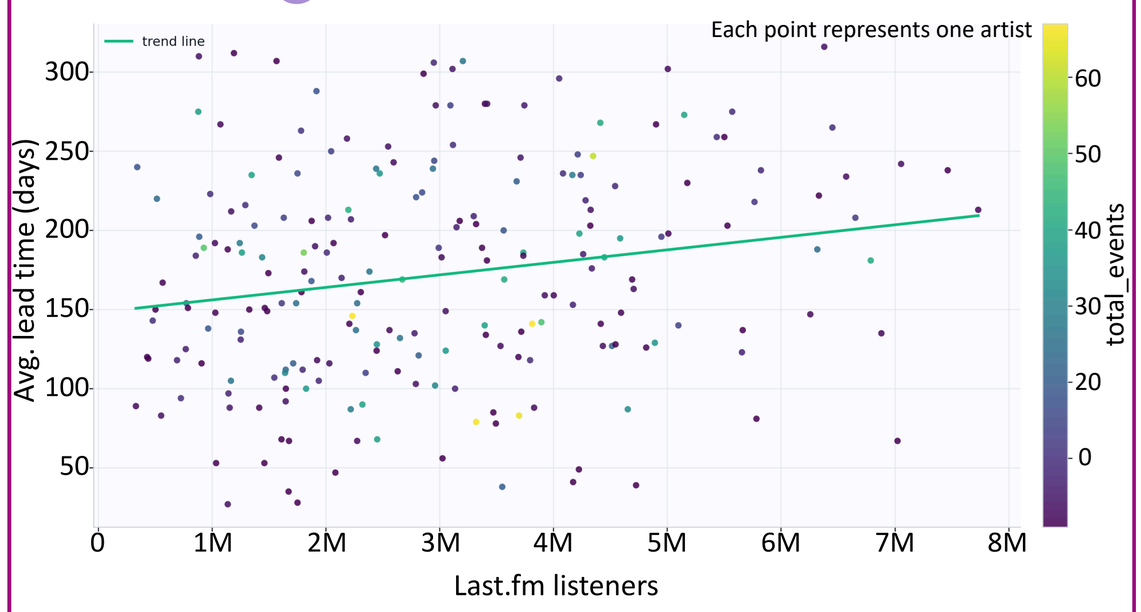
How do artists plan their tours and is the structure of concert scheduling related to their digital popularity?

Artist Popularity vs. Weekend Scheduling



The share of concerts scheduled on weekends varies across popularity groups without a clear pattern. Both more and less popular artists show similar distributions of weekend performances. This suggests that scheduling decisions are not strongly driven by audience size alone.

Listeners vs. Lead Time



There is a slight relationship between listener count and the time between ticket sale and concert date. Lead times tend to increase with popularity, but vary widely across artists. This indicates that tour planning is influenced not only by demand, but also by factors.

Bibliography and References

Last.fm API – Listener statistics and artist popularity data | https://www.last.fm/api
Ticketmaster Discovery API – Concert and event data | https://developer.ticketmaster.com/
RestCountries API – Country and geographic metadata | https://restcountries.com/
Spotify Chart Data | https://charts.spotify.com/

Our Website: https://datascience-project-rchwdmnkmtes77pdt4rhhx.streamlit.app

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